

MySQL Exploratory Data Analysis

EXPLORATORY DATA ANALYSIS PROCESS



Project Report

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Project Type: Exploratory Data Analysis

Database System: MySQL 8.x

Development Tool: MySQL Workbench

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1. Project Overview

This project presents an Exploratory Data Analysis, or EDA, of sales-related data using MySQL. The dataset contains information about customers, products, and sales transactions.

The main purpose of the project was to understand the available data and answer useful business questions. The analysis focused on customer characteristics, product structure, sales measures, geographical performance, time-based activity, and rankings.

The project uses three main tables:

- `dim_customers`
- `dim_products`
- `fact_sales`

Table	Description
<i>dim_customers</i>	Customer details such as name, country, gender, marital status, and birthdate.
<i>dim_products</i>	Product information including categories, subcategories, product lines, and costs.
<i>fact_sales</i>	Transaction details such as order number, customer key, product key, dates, sales amount, quantity, and price.

2. Project Objectives

The main objectives of this project were to:

- Understand the structure of the database.
- Examine the quality and completeness of the data.
- Explore important customer and product dimensions.
- Identify the time period covered by the sales records.
- Calculate the main business measures.
- Compare sales performance across different groups.
- Identify high-performing and low-performing entities.
- Present useful findings in a clear and understandable form.
- Practice SQL concepts in a real data analysis scenario.

EXPLORATORY DATA ANALYSIS PROCESS

3. Database Exploration

Database exploration was the first stage of the project. Its purpose was to understand how the data was organized before performing calculations.

This stage examined the tables available in the database, their columns, data types, and total number of records. Small samples from the tables were also reviewed to understand the actual format of the stored information.

Database exploration was necessary because analysis should not begin without understanding the structure of the data. It helped identify which table contained descriptive information and which table contained measurable sales activity.

The three tables follow a simple dimensional structure:

- dim_customers describes the customers.
- dim_products describes the products.
- fact_sales records sales transactions.

The customer and product keys connect the sales table with the two dimension tables. These relationships made it possible to analyze revenue and quantity using customer or product information.

4. Data Quality Checks

Basic data-quality checks were performed before the main analysis. The purpose of this stage was to determine whether the data was suitable for exploration.

The checks focused on:

- Duplicate customer keys
- Duplicate product keys
- Missing values in important sales fields
- Negative sales amounts
- Zero or negative quantities
- Negative prices
- Sales records without matching customers
- Sales records without matching products

These checks were important because incorrect or incomplete records can produce misleading totals and rankings. For example, duplicate keys may repeat information, while unmatched customer or product keys may prevent records from appearing correctly in joined results.

The quality checks were used for validation only. The original dataset was not changed during the EDA process.

5. Dimensions Exploration

Dimension exploration was used to understand the descriptive categories available in the data.

The customer dimensions included:

- Country
- Gender
- Marital status

The product dimensions included:

- Category
- Subcategory
- Product name
- Product line
- Maintenance classification

This stage showed how customers and products could be grouped during later analysis. For example, customers could be compared by country, while products could be compared by category or subcategory.

Dimension exploration also helped identify the level of detail available in the dataset. A product could be studied individually or as part of a broader category. Similarly, customer activity could be examined at an individual or geographical level.

6. Date Exploration

Date exploration was performed to understand the time coverage of the dataset.

The earliest and latest order dates were identified, and the total sales period was calculated in months. This provided a clear view of how much historical activity was available for analysis.

Customer birthdates were also explored to determine the oldest and youngest customer ages in the dataset.

Sales were then summarized by year and month. These summaries made it possible to observe how revenue, quantity, and order activity changed over time.

Date exploration is an important part of EDA because totals alone do not show whether business activity is increasing, decreasing, or changing between periods.

Note: Customer ages in this project are calculated using the current date. Therefore, the displayed ages may change when the query is executed in the future.

7. Measures Exploration

Measures exploration focused on the main numeric indicators of the business.

The following measures were calculated:

- Total sales revenue
- Total quantity sold
- Average recorded price
- Number of unique orders
- Number of available products
- Number of registered customers
- Number of purchasing customers
- Average order value

These measures provide a general overview of the scale of the business activity contained in the dataset.

The difference between registered customers and purchasing customers was also considered. Registered customers represent all customer records, while purchasing customers are those connected to at least one sales transaction.

Average order value was calculated by dividing total sales revenue by the number of unique orders. This measure indicates the average revenue generated by an order.

The measures provide a useful summary, but they do not explain where the results came from. For that reason, the next stages divided the measures across customers, products, categories, countries, and time periods.

8. Magnitude Analysis

Magnitude analysis was used to compare the size of different groups.

This stage examined:

- Number of customers by country
- Number of customers by gender
- Number of products by category
- Average product cost by category
- Revenue by product category
- Revenue and quantity by country
- Orders and revenue by customer

Magnitude analysis helped show how business activity was distributed. Instead of viewing a single total, the analysis divided that total across meaningful dimensions.

For example, revenue by category showed which product groups made the largest contribution to sales. Revenue by country showed how sales were distributed geographically. Customer-level analysis showed how much revenue and how many orders were connected to individual customers.

The purpose was to compare groups objectively. The analysis shows differences in the available data, but it does not automatically explain the cause of those differences.

9. Ranking Analysis

Ranking analysis was performed to identify entities at the top and bottom of selected measures.

The project identified:

- Five products with the highest revenue
- Five sold products with the lowest revenue
- Ten customers with the highest revenue
- Three customers with the fewest orders
- Revenue positions of products using a window function

Ranking is useful because a full list of customers or products can be difficult to interpret. A smaller top or bottom group makes important results easier to understand.

The project used both `LIMIT` and the `DENSE_RANK()` window function. `LIMIT` returned a fixed number of rows, while `DENSE_RANK()` assigned positions according to product revenue.

The phrase lowest-revenue products was used instead of worst-performing products. Low revenue does not always mean poor performance. A product may be new, temporarily unavailable, or intended for a smaller market. The dataset does not provide enough information to determine those causes.

10. SQL Skills Applied

This project demonstrates practical use of the following SQL concepts:

● SELECT	● UNION ALL
● DISTINCT	● Common Table Expressions
● WHERE	● Window functions
● ORDER BY	● Conditional checks
● GROUP BY	● Date functions
● HAVING	● Null handling
● Aggregate functions	● Top and bottom ranking
● INNER JOIN	● Database metadata exploration
● LEFT JOIN	

11. Main Findings

Instructions: The section below should contain only results produced by the final MySQL queries. Replace the highlighted bracketed fields with the actual values before submitting or sharing this report.

Database and Data Quality

- The database contained **[customer row count]** customer records, **[product row count]** product records, and **[sales row count]** sales records.
- The duplicate-key checks returned **[number]** duplicate customer keys and **[number]** duplicate product keys.
- The sales table contained **[number]** records with missing values in important fields.
- The relationship checks found **[number]** unmatched customer records and **[number]** unmatched product records.

Note: If a check returned no rows or a count of zero, state that no problem was found by that specific check.

Sales Period

- The first recorded order date was **[first order date]**.
- The last recorded order date was **[last order date]**.
- The dataset covered approximately **[number]** months of sales activity.

Customer Findings

- The dataset contained **[total customers]** registered customers.
- **[purchasing customers]** customers were connected to at least one order.
- **[country]** had the largest number of registered customers.
- The highest-revenue customer was **[customer name or customer key]**, with total revenue of **[value]**.

Note: Use customer keys instead of customer names in the public report if you prefer to reduce personal information.

Product Findings

- The dataset contained **[total products]** unique products.
- **[category]** contained the largest number of products.
- **[category]** had the highest average product cost.
- **[product name]** generated the highest product revenue.
- **[product name]** generated the lowest revenue among products with recorded sales.

Revenue and Order Findings

- Total sales revenue was **[total revenue]**.
- A total of **[total quantity]** units were sold.
- The dataset contained **[total orders]** unique orders.
- The average recorded price was **[average price]**.
- The average revenue per order was **[average order value]**.

Category and Geographical Findings

- **[category]** generated the highest category revenue.
- **[country]** generated the highest geographical revenue.
- **[country]** had the highest quantity of sold items.

Time-Based Findings

- The year with the highest revenue was **[year]**.
- The month with the highest revenue was **[year-month]**.
- Sales **[increased/decreased/fluctuated]** across the available time period.

Note: Use the last statement only after checking the yearly and monthly results. Do not state that sales increased if the figures actually decreased or changed irregularly.

12. Conclusion

This project used MySQL to perform a structured exploratory analysis of customer, product, and sales data.

The analysis began by examining the database structure and checking basic data quality. It then explored customer and product dimensions, reviewed the available date range, calculated important business measures, compared the magnitude of different groups, and ranked selected products and customers.

The project demonstrates how SQL can transform raw relational data into understandable business information. It also shows that EDA includes both exploration and analysis. Exploration explains what data is available, while analysis helps identify patterns, differences, and important results.

The findings provide an initial understanding of the dataset. They should not be treated as explanations of why a particular result occurred because the available tables do not contain information about marketing activity, inventory shortages, discounts, customer satisfaction, economic conditions, or other possible causes.